

The SHA-256 Die: A Formal Theory of Causality, Closure, and Manifold

Seven-Level Formalization of the 64-Cell SHA-256 Recurrence over $(\mathbb{Z}/2^{32}\mathbb{Z})^8$

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Abstract

This thesis develops a complete seven-level formal theory of the SHA-256 compression function as a fixed 64-cell nonlinear recurrence over the module $(\mathbb{Z}/2^{32}\mathbb{Z})^8$, parametrised by a displacement field carrying the message schedule. The central analytical device is the NOP backbone: the eternal orbit of the recurrence when its displacement field is zeroed. Any real message block is treated as a perturbation field superimposed on this backbone.

Three sharp structural invariants are established. The ground witness: the NOP backbone evaluates its ground-fold operator $G(x) = \text{Sigma}_0(a) + \text{Maj}(a,b,c)$ to the exact constant `0x08909ae5` at round zero, an absolute message-independent coordinate of the die's state space. The word-level support diameter $D_{\text{word}} = 4$: a single-word perturbation in W_0 saturates all eight state lanes in exactly four rounds under the Boolean support model governed by the 8x8 lane-dependency matrix M . The bit-level support diameter $D_{\text{bit}} = 6$: a single perturbed bit at position j saturates all 256 state bits in at most six rounds, with the exact radius $\rho(j)$ stratified as 4 for $j = 0, 5$ for $1 \leq j \leq 25$, and 6 for $26 \leq j \leq 31$.

Beyond support closure, the theory is extended through four additional levels: exact carry realization via a bit-level automaton; seam geometry tracking the exact perturbation skeleton at the two active injection points; closure functionals measuring age-class equalization; and the residual smoothing band — after round 6, the die enters a stable density oscillation centered near $16 = 32/2$, continuing to round 64.

A dual-pipeline topology is identified: the eight-register state decomposes into two parallel four-register shift chains with complementary chirality, connected by a cross-coupling and jointly activated by the injection vector $b = [1, 0, 0, 0, 1, 0, 0, 0]^T$. This structure is isomorphic to the bipolar junction transistor. The manifold interpretation unifies the framework: the die is one lawful closure shape recurring through many different message streams.

Keywords: SHA-256, nonlinear recurrence, Boolean support, carry propagation, support diameter, dual-pipeline, age-weight law, closure functionals, manifold.

Chapter 1 — Introduction

1.1 Motivation

SHA-256 is one of the most widely deployed cryptographic primitives in the world. Despite this ubiquity, its internal structure as a dynamical system has received little formal treatment at the level of exact invariants and phase analysis. The algorithm is typically treated as an opaque one-way function — studied for what it does rather than for what it is as a mathematical object.

This thesis proposes the die interpretation. SHA-256 is modelled as a formal 64-cell nonlinear recurrence on $(\mathbb{Z}/2^{32} \mathbb{Z})^8$, equipped with fixed structural rails and a variable displacement field. The word die evokes the casting die of manufacture: a fixed mould whose geometry determines what shapes can be produced from it.

1.2 The NOP Backbone

The central analytical move is the identification of the NOP backbone: the orbit of the recurrence when its message schedule is identically zero. This backbone depends only on the fixed initialisation vector H_0 and the fixed round constants K . It is the machine running without a user.

Any real message block is treated as a perturbation field $W = (W_0, \dots, W_{63})$ superimposed on this backbone. The round-zero identity $T_{1,0} - T_1^{\wedge}(0)_0 = W_0$ shows that at round zero the message enters cleanly and linearly. After round zero, the states diverge through 64 rounds of nonlinear recursion.

1.3 The Seven Levels

The theory develops in seven nested levels, each strictly refining the last:

- Level 0: Scalar ground invariant $T_2^{\wedge}(0)_0 = 0x08909ae5$
- Level 1: Word-support transport via 8x8 matrix M , diameter $D_{\text{word}} = 4$
- Level 2: Bit-support transport via 256-lane map Ψ , diameter $D_{\text{bit}} = 6$ with exact radius profile $\rho(j)$
- Level 3: Exact carry realization via bit-level automaton $C(x, \delta)$
- Level 4: Seam geometry — exact perturbation skeletons at rounds 3 and 4
- Level 5: Closure functionals — three scalar measures of age-class equalization
- Level 6: Residual smoothing band — density oscillates near $16 = 32/2$ for rounds 7 through 64

Chapter 2 — State Space and Round Map

2.1 The Die

Definition 2.1.

The SHA-256 die is the dynamical system $D = (X, \{\Phi_r\}, H_0, K, W)$ where $X = (\mathbb{Z}/2^{32} \mathbb{Z})^8$ is the state space with elements written as 8-word column vectors $x = (a, b, c, d, e, f, g, h)^T$; Φ_r is the round map; H_0 is the fixed initialisation vector; $K = (K_0, \dots, K_{63})$ are the round constants; and $W = (W_0, \dots, W_{63})$ is the message schedule. The initial condition is $x_0 = H_0$. The die evolves by $x_{r+1} = \Phi_r(x_r, W_r)$ for $r = 0, \dots, 63$.

2.2 Round Operators

Definition 2.2.

The right-rotation operator $\text{ROTR}^n(x)$ rotates the 32-bit word x right by n positions. The SHA-256 sigma operators are:

$$\text{Sigma}_0(x) = \text{ROTR}^{22}(x) \text{ XOR } \text{ROTR}^{13}(x) \text{ XOR } \text{ROTR}^{3}(x)$$

$$\text{Sigma}_1(x) = \text{ROTR}^{6}(x) \text{ XOR } \text{ROTR}^{11}(x) \text{ XOR } \text{ROTR}^{25}(x)$$

The nonlinear Boolean gates are:

$$\text{Ch}(e, f, g) = (e \text{ AND } f) \text{ XOR } (\text{NOT } e \text{ AND } g)$$

$$\text{Maj}(a, b, c) = (a \text{ AND } b) \text{ XOR } (a \text{ AND } c) \text{ XOR } (b \text{ AND } c)$$

Ch and Maj act bitwise: they operate independently on each bit position. Sigma_0 and Sigma_1 are global: they couple bit positions through the three rotation offsets.

2.3 The Round Weights and State Update

The two weight scalars at round r are:

$$T1_r = h_r + \text{Sigma}_1(e_r) + \text{Ch}(e_r, f_r, g_r) + K_r + W_r$$

$$T2_r = \text{Sigma}_0(a_r) + \text{Maj}(a_r, b_r, c_r)$$

All arithmetic is modulo 2^{32} . The state update is:

$$a_{r+1} = T1_r + T2_r$$

$$e_{r+1} = d_r + T1_r$$

$$b_{r+1} = a_r, \quad c_{r+1} = b_r, \quad d_{r+1} = c_r$$

$$f_{r+1} = e_r, \quad g_{r+1} = f_r, \quad h_{r+1} = g_r$$

Six of the eight state words are pure register shifts. Only a and e receive nonlinear injections at each round. This structural sparsity is a fundamental feature of the die.

2.4 The Shift-Injection Decomposition

Proposition 2.1.

Define the 8x8 shift matrix P (with $P_{\{i,i-1\}} = 1$ for $i = 1, \dots, 7$ and all other entries zero) and standard basis vectors $u_a = e_0$, $u_e = e_4$. Then the round map satisfies:

$$x_{\{r+1\}} = P x_r + u_a (T_{1_r} + T_{2_r}) + u_e T_{1_r}$$

2.5 The Prime-Root Rails

H_o consists of the first 32 bits of the fractional parts of the square roots of the first eight primes. K consists of the first 32 bits of the fractional parts of the cube roots of the first 64 primes. Square roots of primes are maximally irrational — their fractional parts are badly approximable by rationals with small denominators — which prevents low-period resonances at startup.

Register	Source	Value
$H_o[0]$ (a)	$\sqrt{2}$	0x6a09e667
$H_o[1]$ (b)	$\sqrt{3}$	0xbb67ae85
$H_o[2]$ (c)	$\sqrt{5}$	0x3c6ef372
$H_o[3]$ (d)	$\sqrt{7}$	0xa54ff53a
$H_o[4]$ (e)	$\sqrt{11}$	0x510e527f
$H_o[5]$ (f)	$\sqrt{13}$	0x9b05688c
$H_o[6]$ (g)	$\sqrt{17}$	0x1f83d9ab
$H_o[7]$ (h)	$\sqrt{19}$	0x5be0cd19

Chapter 3 — The NOP Backbone and Ground Plane

3.1 The NOP Manifold

Definition 3.1.

The NOP backbone of the SHA-256 die is the trajectory $\{x^{(o)}_r\}$ defined by $x^{(o)}_0 = H_o$ and $x^{(o)}_{\{r+1\}} = \Phi_r(x^{(o)}_r, o)$. It is the orbit of the die when the displacement field is identically zero.

For any real message block, the trajectory decomposes as $x_r = x^{(o)}_r + \delta x_r$ in $(Z/2^{32}Z)^8$, where δx_r is the perturbation vector. The initial perturbation is $\delta x_0 = o$.

3.2 The Ground Witness

Definition 3.2.

The ground-fold operator is $G: (Z/2^{32}Z)^8 \rightarrow Z/2^{32}Z$ defined by $G(x) = \text{Sigma}_o(a) + \text{Maj}(a, b, c)$. Note $G(x_r) = T_{2_r}$.

Theorem 3.1 (Ground Witness).

The ground-fold operator evaluated at the NOP initial state satisfies:

$$T_2^{(o)} = G(H_o) = \text{Sigma}_o(H_o[0]) + \text{Maj}(H_o[0], H_o[1], H_o[2]) = \text{oxo8909ae5}$$

Proof: Direct computation using $H_o[0] = \text{ox6a0ge667}$, $H_o[1] = \text{oxbb67ae85}$, $H_o[2] = \text{ox3c6ef372}$. The three rotations of ox6a0ge667 are XORed to produce Sigma_o ; the bitwise majority of the three values produces Maj ; their modular sum is oxo8909ae5 . Verification code:

```
def rotr(x,n): return ((x>>n)|(x<<(32-n)))&0xFFFFFFFF sigmao = lambda x: rotr(x,2)^rotr(x,13)^rotr(x,22) maj = lambda a,b,c: (a&b)^(a&c)^(b&c) result = (sigmao(ox6a0ge667) + maj(ox6a0ge667,oxbb67ae85,ox3c6ef372)) % 2**32 # result == oxo8909ae5
```

The ground witness oxo8909ae5 is an absolute structural invariant of SHA-256. It is the value of the backbone's ground-fold register before any message information enters the system.

3.3 The Round-Zero Perturbation Identity

Theorem 3.2.

For any message word W_o , the perturbation at round o satisfies $T_{1_o} - T_1^{(o)} = W_o$ and $T_{2_o} = T_2^{(o)}$ (T_2 is blind to W_o at round o). Therefore:

$$\Delta a_1 = \Delta e_1 = W_o$$

The message enters only two words on the first step. The entire round- o perturbation lives in $\Delta T_{1_o} = W_o$; the ground fold T_2 does not move.

The exact NOP backbone values through round 4:

Round r	$a^{(o)}_r$	$e^{(o)}_r$
1	oxfco8884d	ox98c7e2a2
2	ox7ad96290	ox9df1b216
3	oxf3dd6c3f	oxc57b68fb
4	ox0a24b1aa	ox909cf5c9

Chapter 4 — The Dual-Pipeline Topology

4.1 Two Shift Chains

Definition 4.1.

The a -chain is the ordered sequence (a, b, c, d) , satisfying $b_{r+1} = a_r$, $c_{r+1} = b_r$, $d_{r+1} = c_r$. The e -chain is (e, f, g, h) , satisfying $f_{r+1} = e_r$, $g_{r+1} = f_r$, $h_{r+1} = g_r$. Values propagate rightward toward the tail by one position per round. Each chain head (a and e) receives a nonlinear injection each round.

4.2 Chirality

Definition 4.2.

The a-chain has present-tense chirality: $T2_r = \text{Sigma}_o(a_r) + \text{Maj}(a_r, b_r, c_r)$ reads from $\{a_r, b_r, c_r\}$, the three most recently updated positions of the a-chain. The e-chain has past-tense chirality: $T1_r$ reads from $\{h_r, e_r, f_r, g_r\}$, the full e-chain including $h_r = e_{\{r-3\}}$ (three rounds old). $T2$ measures recent change; $T1$ integrates a longer history.

4.3 Injection Vector

Definition 4.3.

The message injection vector b in $\{0,1\}^8$ is $b = [1,0,0,0,1,0,0,0]^T$. Both pipeline heads are activated simultaneously by each message word W_r . The injection is orthogonal to the shift direction — the message does not travel on the same axis as the register flow.

4.4 Cross-Coupling

Definition 4.4.

The cross-coupling $d_r \rightarrow e_{\{r+1\}}$ via the equation $e_{\{r+1\}} = d_r + T1_r$ connects the tail of the a-chain to the head of the e-chain. The value of a from three rounds ago (encoded in $d_r = a_{\{r-3\}}$) contributes to the next value of e , coupling both chains' histories.

4.5 Transistor Correspondence

The dual-pipeline topology is structurally isomorphic to the bipolar junction transistor. The e-chain (reading $T1$, past-tense, accumulated history) corresponds to the Emitter. The a-chain head (receiving $T1 + T2$, resolved injection) corresponds to the Collector. The injection vector b (entering both heads simultaneously, orthogonal to shift direction) corresponds to the Base. The cross-coupling $d_r \rightarrow e_{\{r+1\}}$ is the substrate feedback. NPN chirality corresponds to the a-chain (present-tense active flow); PNP chirality to the e-chain (past-tense accumulated reading).

The transistor is not a component with three wires. It is a single closure seam where three lawful paths meet simultaneously: the offer (Emitter), the admission (Base), and the resolved intake (Collector). The decision dimension is always orthogonal to the transport dimension. The die encodes this geometrically: the injection vector b is orthogonal to the shift matrix P in the sense that $Pu_a = 0$ (the first row of P is zero).

Chapter 5 — Word-Level Support Transport

5.1 The Boolean Support Model

Definition 5.1.

Given perturbation trajectory $\{\delta x_r\}$, the word-support indicator at round r is σ_r in $\{0,1\}^8$ where $(\sigma_r)_j = 1$ if $(\delta x_r)_j$ is nonzero. The Boolean support model tracks which words are affected, not their exact values. This is the worst-case (largest possible) diffusion estimate.

5.2 The Lane-Dependency Matrix

Proposition 5.1.

The 8x8 Boolean lane-dependency matrix M of the SHA-256 die, with $M_{ij} = 1$ meaning lane i at round $r+1$ depends on lane j at round r , is:

$$M = \begin{matrix} & \text{row a:} & [1\ 1\ 1\ 0\ 1\ 1\ 1\ 1] & (\text{T2 reads } \{a,b,c\}; \text{ T1 reads } \{e,f,g,h\}) \\ & \text{row b:} & [1\ 0\ 0\ 0\ 0\ 0\ 0\ 0] & (b_{r+1} = a_r) \\ & \text{row c:} & [0\ 1\ 0\ 0\ 0\ 0\ 0\ 0] & (c_{r+1} = b_r) \\ & \text{row d:} & [0\ 0\ 1\ 0\ 0\ 0\ 0\ 0] & (d_{r+1} = c_r) \\ & \text{row e:} & [0\ 0\ 0\ 1\ 1\ 1\ 1\ 1] & (e_{r+1} = d_r + T1_r; \text{ T1 reads } \{e,f,g,h\}) \\ & \text{row f:} & [0\ 0\ 0\ 0\ 1\ 0\ 0\ 0] & (f_{r+1} = e_r) \\ & \text{row g:} & [0\ 0\ 0\ 0\ 0\ 1\ 0\ 0] & (g_{r+1} = f_r) \\ & \text{row h:} & [0\ 0\ 0\ 0\ 0\ 0\ 1\ 0] & (h_{r+1} = g_r) \end{matrix}$$

5.3 Theorem: $D_{\text{word}} = 4$

Theorem 5.1.

For single injection at round 0, the word-level support diameter $D_{\text{word}} = \min\{n \geq 1 : \sigma_n = 1\} = 4$.

Proof: Explicit computation of $M^{[n-1]} * b$. The support orbit is:

Round r	Support set	Count
1 (injection)	$\{a, e\}$	2
2	$\{a, b, e, f\}$	4
3	$\{a, b, c, e, f, g\}$	6
4	All: $\{a, b, c, d, e, f, g, h\}$	8

The support grows by exactly two lanes per round, one per chain, reflecting the symmetric dual-pipeline structure. $D_{\text{word}} = 4$ is the first round at which $\sigma_r = 1$. The growth pattern 2, 4, 6, 8 is forced by the geometry: the a-chain fills b, c, d in rounds 2, 3, 4 while the e-chain fills f, g, h simultaneously.

Chapter 6 — The 256-Lane Bit-Support State

6.1 Bit-Support Vectors

Definition 6.1.

For each word w in $\{a, \dots, h\}$ and round r , the bit-support vector $s_{\{w,r\}}$ in $\{0,1\}^{32}$ has $(s_{\{w,r\}})_i = 1$ if bit i of the perturbation $(\delta x_r)_w$ is potentially nonzero. The full 256-lane support state is $\eta_r = (s_a, s_b, s_c, s_d, s_e, s_f, s_g, s_h)^T$ in $\{0,1\}^{256}$.

6.2 Rotation Support Operators

Definition 6.2.

For n in $\{0, \dots, 31\}$, the rotation support operator R_n is the 32×32 permutation matrix with $(R_n s)_i = s_{\{(i+n) \bmod 32\}}$. The Boolean support versions of the sigma operators are:

$$\hat{\text{Sigma}}_0 = R_2 \text{ OR } R_{13} \text{ OR } R_{22}$$

$$\hat{\text{Sigma}}_1 = R_6 \text{ OR } R_{11} \text{ OR } R_{25}$$

These correctly over-approximate: $\text{supp}(\text{Sigma}_o(\delta x))$ is contained in $\hat{\text{Sigma}}_o * \text{supp}(\delta x)$. Both operators are 3-regular circulants; neither is privileged by density before coupling to carry.

6.3 Weight Support Vectors

Definition 6.3.

The bit-support of $T1_r$:

$$\tau^{(1)}_r = s_{\{h,r\}} \text{ OR } \hat{\text{Sigma}}_1 * s_{\{e,r\}} \text{ OR } s_{\{e,r\}} \text{ OR } s_{\{f,r\}} \text{ OR } s_{\{g,r\}} \text{ OR } \omega_r$$

The bit-support of $T2_r$:

$$\tau^{(2)}_r = \hat{\text{Sigma}}_0 * s_{\{a,r\}} \text{ OR } s_{\{a,r\}} \text{ OR } s_{\{b,r\}} \text{ OR } s_{\{c,r\}}$$

$\tau^{(1)}$ reads from the e-chain (past-tense). $\tau^{(2)}$ reads from the a-chain (present-tense). The chirality structure identified at the word level recurs exactly at the bit level.

Chapter 7 — The Carry-Closure Kernel

7.1 Carry Propagation

A carry generated at bit position j in a modular addition propagates upward to positions $j+1, j+2, \dots$, potentially reaching position 31. This process is nonlocal and directional: it propagates upward only.

7.2 The Operator L_{32}

Definition 7.1.

The carry-closure kernel is the 32×32 lower-triangular prefix operator L_{32} defined by $(L_{32} x)_i = \text{OR}\{x_j : j \leq i\}$ for $i = 0, \dots, 31$. Applied to a support vector s , $(L_{32} s)_i = 1$ if any bit $j \leq i$ of s is 1. This captures worst-case carry propagation.

Remark 7.1.

For a perturbation at a single bit j : $L_{32}(\{j\}) = \{j, j+1, \dots, 31\}$, a set of size $32 - j$. A perturbation at the LSB ($j=0$) potentially affects all 32 bits. A perturbation at the MSB ($j=31$) affects only itself. This directionality is the root cause of the $\rho(j)$ stratification.

7.3 The 256-Lane Update Rule

Theorem 7.1.

The bit-level support state evolves according to the map Ψ :

$$\begin{aligned} s_{\{a,r+1\}} &= L_{32}(\tau^{(1)}_r \text{ OR } \tau^{(2)}_r) \\ s_{\{e,r+1\}} &= L_{32}(s_{\{d,r\}} \text{ OR } \tau^{(1)}_r) \\ s_{\{b,r+1\}} &= s_{\{a,r\}}, \quad s_{\{c,r+1\}} = s_{\{b,r\}}, \quad s_{\{d,r+1\}} = s_{\{c,r\}} \\ s_{\{f,r+1\}} &= s_{\{e,r\}}, \quad s_{\{g,r+1\}} = s_{\{f,r\}}, \quad s_{\{h,r+1\}} = s_{\{g,r\}} \\ \eta_{r+1} &= \Psi(\eta_r, \omega_r) \end{aligned}$$

Chapter 8 — Bit-Level Support Diameter

8.1 Single-Bit Injection

Proposition 8.1.

Under single-bit injection at position j , the round-1 bit support is $s_{\{a,1\}} = s_{\{e,1\}} = L_{32}(e_j) = \{j, j+1, \dots, 31\}$. The initial support has size $32 - j$. All other round-1 bit-support vectors are zero.

8.2 The Radius Profile

Theorem 8.1 (Bit-Support Radius Profile).

For single-bit injection at position j , the bit-support radius $\rho(j)$ — the first round at which all 256 state bits are in support — is:

$$\begin{aligned} \rho(j) &= 4 \quad \text{for } j = 0 \quad (\text{LSB: } L_{32}(e_0) = 1, \text{ both heads immediately full}) \\ \rho(j) &= 5 \quad \text{for } 1 \leq j \leq 25 \quad (\text{scatter-and-close via rotation in one extra round}) \\ \rho(j) &= 6 \quad \text{for } 26 \leq j \leq 31 \quad (\text{minimal carry potential, two extra rounds required}) \\ D_{\text{bit}} &= \max_j \rho(j) = 6 \end{aligned}$$

8.3 The Carry Excess

Theorem 8.2.

The excess $D_{\text{bit}} - D_{\text{word}} = 6 - 4 = 2$ arises entirely from the directionality of L_{32} . High-order injected bits require two extra rounds: one for $\hat{\Sigma}_0$ or $\hat{\Sigma}_1$ to scatter support to low-order positions, and one for L_{32} to close from those positions to all remaining bits. If L_{32} were isotropic (acting on both high and low bits simultaneously) the excess would be zero.

Bit position j	Initial support size	$\rho(j)$	Limiting factor
$j = 0$ (LSB)	32 (full)	4	Word-level geometry only

Bit position j	Initial support size	$\rho(j)$	Limiting factor
j in $[1, 25]$	7 to 31	5	One round: rotation scatter
j in $[26, 31]$	1 to 6	6	Two rounds: carry kernel limit

Chapter 9 — Exact Carry Realization

9.1 The Carry Automaton

Definition 9.1.

For exact addition $y = x + \delta \pmod{2^{32}}$, define the carry sequence:

$$c_{-1} = 0$$

$$c_i = (x_i \text{ AND } \delta_i) \text{ OR } (x_i \text{ AND } c_{i-1}) \text{ OR } (\delta_i \text{ AND } c_{i-1}) \text{ for } i = 0, \dots, 31$$

$$y_i = x_i \text{ XOR } \delta_i \text{ XOR } c_{i-1}$$

The exact changed-bit indicator is $\Delta_i(x, \delta) = x_i \text{ XOR } y_i = \delta_i \text{ XOR } c_{i-1}$.

Theorem 9.1.

For one-hot injection $\delta = 2^j$ the indicator simplifies to: $\Delta_i = 0$ for $i < j$; $\Delta_i = 1$ for $i = j$; $\Delta_i = \text{product}\{x_t : t = j, \dots, i-1\}$ for $i > j$. The exact changed-bit set is $C_x(j) = \{j, \dots, m_x(j)\}$ where $m_x(j) = \min\{i \geq j : x_i = 0\}$. The carry-span length is $\lambda_x(j) = m_x(j) - j + 1$.

9.2 Exact Round-1 Carry Spans

Using NOP baselines $a^{(0)}_1 = \text{oxfc08884d}$ and $e^{(0)}_1 = \text{oxg8c7e2a2}$:

a-seam carry spans $\lambda_a(j)$ for $j = 0, \dots, 31$:

$$(2, 1, 3, 2, 1, 1, 2, 1, 1, 1, 3, 2, 1, 1, 1, 4, 3, 2, 1, 1, 1, 3, 2, 1, 1, 1, 6, 5, 4, 3, 2, 1)$$

e-seam carry spans $\lambda_e(j)$ for $j = 0, \dots, 31$:

$$(1, 2, 1, 1, 1, 2, 1, 2, 1, 2, 1, 1, 1, 1, 4, 3, 2, 1, 6, 5, 4, 3, 2, 1, 1, 1, 1, 3, 2, 1, 1, 1)$$

The two seams are injection-symmetric at round 0 but not carry-symmetric: different NOP baselines produce different carry-span distributions. This is the first point where the two pipelines diverge in exact dynamics.

Chapter 10 — Seam Geometry

10.1 Exact Perturbation Skeletons

Theorem 10.1 (Round-3 Skeleton).

For single-bit injection $W_o = 2^j$:

$$\Delta x_3 = (\Delta a_3, \Delta a_2, 2^j, 0, \Delta e_3, \Delta e_2, 2^j, 0)$$

where $\Delta e_3 = \Delta T_{1_2}$ and $\Delta a_3 = \Delta T_{1_2} + \Delta T_{2_2} \pmod{2^{32}}$. The seed 2^j is still explicitly visible in positions c and g.

Theorem 10.2 (Round-4 Skeleton).

For single-bit injection $W_o = 2^j$:

$$\Delta x_4 = (\Delta a_4, \Delta a_3, \Delta a_2, 2^j, \Delta e_4, \Delta e_3, \Delta e_2, 2^j)$$

where $\Delta e_4 = \Delta T_{1_3}$ and $\Delta a_4 = \Delta T_{1_3} + \Delta T_{2_3} \pmod{2^{32}}$. Round 4 is the first layer with full word support but the seed 2^j remains visible in the tail lanes d and h.

10.2 Hamming Weight Ranges

Lane	Round 3 range	Round 4 range	Notes
Δa	[13, 21]	[12, 20]	Active seam (a-chain head)
Δe	[7, 21]	[11, 21]	Active seam (e-chain head)
Δb	—	[13, 21]	= round-3 a-seam
Δc	—	[7, 19]	= round-2 a-seam
Δd	—	[1, 6]	= carry span of 2^j only
Δf	—	[7, 21]	= round-3 e-seam
Δg	—	[3, 16]	= round-2 e-seam
Δh	—	[1, 7]	= carry span of 2^j only

The tail lanes d and h carry only the carry span of the original seed bit 2^j at round 4. This confirms that round 4 is topological acceptance, not bit-density closure. The word fabric is full, but the density distribution is still stratified by age.

Chapter 11 — The Age-Weight Law and Closure Functionals

11.1 Age Classes

Definition 11.1.

Define three age classes: head lanes $H_r = \{a_r, e_r\}$ (most recently injected); mid lanes $M_r = \{b_r, c_r, f_r, g_r\}$ (one to two rounds old); tail lanes $T_r = \{d_r, h_r\}$ (three rounds old). For one-hot injections W_o

$= 2^j$ over $j = 0, \dots, 31$, define class mean Hamming weights $\mu_H(r)$, $\mu_M(r)$, $\mu_T(r)$ averaged over the class and all 32 injection positions.

Theorem 11.1 (Age-Weight Law).

Round	μ_H (head)	μ_M (mid)	μ_T (tail)	Age spread
4	15.73	12.93	1.84	13.89
5	15.88	15.73	10.12	5.75
6	15.78	15.80	15.73	0.07

By round 6 the three age classes have equalized to within 0.07 Hamming-weight units. The tail lanes rise from mean weight 1.84 at round 4 to 15.73 at round 6 — a gain of 13.89 units in two rounds.

11.2 Closure Functionals

Definition 11.2 (Age-Spread Functional).

$$E_{\text{age}}(r) = \max\{\mu_H, \mu_M, \mu_T\}(r) - \min\{\mu_H, \mu_M, \mu_T\}(r)$$

Definition 11.3 (Lane-Variance Functional).

$$V(r) = (1/8) * \sum_L [\mu_L(r) - \mu_{\text{bar}}(r)]^2$$

Definition 11.4 (Lane-Range Functional).

$$R(r) = \max_L \mu_L(r) - \min_L \mu_L(r)$$

Functional	Round 4	Round 5	Round 6	Meaning at zero
$E_{\text{age}}(r)$	13.89	5.75	0.07	Age classes equalized
$V(r)$	33.67	7.36	0.41	Lane densities equalized
$R(r)$	15.22	8.78	2.41	No lane outliers remaining

Chapter 12 — The Four-Phase Law and Residual Smoothing

12.1 The Four-Phase Law

Theorem 12.1 (Four-Phase Law of the Die).

The SHA-256 die's response to a single-word perturbation W_0 proceeds through exactly four phases:

Phase I — Injection ($r = 0$ to 3). The perturbation is visibly tied to the seed and its immediate descendants. Only a and e , then a, b, e, f , then a, b, c, e, f, g are occupied sequentially.

Phase II — Acceptance ($r = 4$). All eight lanes are occupied. $D_{\text{word}} = 4$ is achieved. Bit-density remains stratified: tail lanes have mean weight ~ 1.84 versus head lanes ~ 15.73 .

Phase III — Closure ($r = 5$ and 6). Bit-density equalizes across age classes. $D_{\text{bit}} = 6$ is achieved. Tail lanes rise from mean 1.84 to 15.73 in two rounds.

Phase IV — Residual Smoothing ($r > 6$). No new support is created. The die rebalances density inside the already-full fabric.

$D_{\text{word}} = 4$ marks topological acceptance

$D_{\text{bit}} = 6$ marks support closure

$r > 6$ is residual smoothing around density ~ 16

12.2 The Residual Smoothing Band

Theorem 12.2.

The global mean perturbation density $\mu_{\text{bar}}(r) = (1/8) * \sum_L \mu_L(r)$ evolves:

Round r	$\mu_{\text{bar}}(r)$
4	10.86
5	14.37
6	15.78

For rounds 6 through 64, the die enters a narrow oscillatory band:

$$15.61 \leq \mu_{\text{bar}}(r) \leq 16.53 \text{ for } r = 6, \dots, 64$$

Minimum at $r = 22$; maximum at $r = 53$. The band is centered near $16 = 32/2$, the half-width of the 32-bit word fabric. The rotation offsets in Sigma_0 and Sigma_1 are designed so that every bit position contributes approximately equally to the three rotated outputs; in the long run each bit has roughly equal probability of being in support, giving mean density near 16.

Chapter 13 — The Complete Formal Stack

13.1 Seven Levels

Level	Object	Key result
0 — Ground witness	Scalar constant $G(H_0)$	0xo8909ae5
1 — Word transport	8x8 Boolean matrix M	$D_{\text{word}} = 4$

Level	Object	Key result
2 — Bit transport	256-lane map Psi	D_bit = 6, rho(j) profile
3 — Exact carry	Bit automaton C(x,delta)	lambda_x(j) exact spans
4 — Seam geometry	Perturbation skeletons	Round 3, 4 exact maps
5 — Closure fns	E_age, V, R, mu_bar	Near-zero by r = 6
6 — Residual band	Density interval	mu_bar in [15.61, 16.53]

13.2 The Key Distinction

Round 4 marks topological acceptance: all eight state words are in support. This is a graph-theoretic fact about the Boolean matrix M — it says nothing about the density of the perturbation within those words.

Round 6 marks support closure: all 256 state bits are in support and density has equalized across age classes. This is both a topological and a metric fact.

Acceptance != Closure != Final Smoothing

13.3 Chirality and Anisotropy

The rotation operators $\hat{\sigma}_0$ and $\hat{\sigma}_1$ are both 3-regular circulants. Neither is privileged by density. The visible asymmetry between the two chains appears only after coupling to lane placement and carry closure. The correct statement is: uniform rotations + lane asymmetry + carry closure = visible chirality of the die. The operators themselves are structurally similar; the asymmetry is in what they read (present-tense versus past-tense values).

Chapter 14 — Geometric Interpretation

14.1 The Glass Key

The NOP backbone defines a natural reference trajectory in state space X. Given the final state x_{64} of a real computation and the NOP final state $x^{(o)}_{64}$, the difference $\Delta x_{64} = x_{64} - x^{(o)}_{64} \pmod{2^{32}}$ encodes the cumulative perturbation after 64 rounds of nonlinear mixing.

Computing Δx_{64} from x_{64} by subtracting the known NOP backbone value is the Glass Key operation. It is a Z-axis read: rather than reading along the computation axis, one reads orthogonally by comparing the real trajectory to the backbone. The Glass Key does not reverse the hash; it reveals the structural footprint of the message on the backbone. The NOP backbone is computable from H_0 and K alone.

14.2 The Manifold Identity

Theorem 14.1 (Manifold Identity).

A manifold is the crossing of two properties:

$M = (\text{many-to-one in flow}) \text{ INTERSECT } (\text{one-to-one in shape})$

Many different message streams W can produce states that fall into the same basin under the die's closure shape (many-to-one in flow). But the closure shape of the die — the NOP backbone, the dual-pipeline topology, the four-phase law — is unique and fixed (one-to-one in shape). Therefore:

the SHA-256 die is one lawful closure shape recurring through many different streams

14.3 Constant Envelope as Ground

SHA-256 always produces a 256-bit output. The envelope length 256 is fixed across all inputs. Therefore the length carries no information — it is the ground condition, not the message. All differential content lives in the internal fold geometry: phase, curvature, chirality, alignment, residue. The ground witness `0xo8g0gae5` is the ground condition of the first round. The NOP backbone is the absolute reference. All perturbations are measured against this reference. The manifold is the shape; the message is the stream passing through it.

14.4 Primitive Grammar

At the bit level, the die operates with three primitives: start (0 to 1 transition), stop (1 to 0 transition), and persist (same state surviving into the next tick). The carry automaton is precisely the mechanism of persist: a carry bit persists through as many consecutive 1-bits as the baseline contains above the injection point. The carry-span length $\lambda_x(j)$ measures exactly how long the persist event lasts.

Chapter 15 — Connections

15.1 Differential Cryptanalysis

The word-level support diameter $D_{\text{word}} = 4$ implies that any differential characteristic on fewer than 4 rounds can potentially exploit lane sparsity. Attacks on 4 or more rounds cannot rely on any lane being clean. The bit-level result refines this: for perturbations at bit positions j in $\{26, \dots, 31\}$, full 256-bit support is not achieved until round 6. A cryptanalyst studying 5-round SHA-256 with high-order bit differences may find residual bit-level sparsity invisible at the word level.

15.2 Keccak Comparison

Property	SHA-256 Die	Keccak-f[1600]
State size	256 bits	1600 bits
Rounds	64	24
Round structure	Sequential dual-pipeline	5 parallel steps
Ground plane	<code>0xo8g0gae5</code> (explicit)	None equivalent
Message injection	Expanded schedule into T_1	Direct XOR into rate
Architecture	Narrow sequential (turbofan)	Wide parallel (scramjet)

15.3 Open Problems

- Exact $\rho(j)$ profile under full modular arithmetic, not just the Boolean support bound
- Fourth-level carry-chain tracking: exact carry bits rather than support
- Extension of the four-phase law to multi-block Merkle-Damgard chains
- Formal proof that the residual band [15.61, 16.53] is tight for all 32 injection positions
- Application of the Glass Key to reduced-round preimage analysis

Chapter 16 — Conclusion

16.1 The Three Invariants

Invariant I — Ground witness: $T_2^{(0)}_0 = \text{0x08909ae5}$

Invariant II — Word-level diameter: $D_{\text{word}} = 4$

Invariant III — Bit-level diameter: $D_{\text{bit}} = 6$

Radius profile: $\rho(j) = 4 / 5 / 6$ by bit position

16.2 The Four-Phase Law

Phase I Injection $r = 0-3$ seeding through pipelines

Phase II Acceptance $r = 4$ D_{word} achieved, density stratified

Phase III Closure $r = 5-6$ D_{bit} achieved, age classes equalized

Phase IV Smoothing $r > 6$ density oscillates near $16 = 32/2$

16.3 The Formal Stack

$(\Phi_r, M, \Psi, L_{32}, C(x, \delta), E_{\text{age}}/V/R, \mu_{\text{bar}} \sim 16)$

16.4 The Manifold

manifold = many-to-one in flow INTERSECT one-to-one in shape

=> there is only one shape, many times

The machine runs without us. Its ground plane was fixed before any silicon was fabricated. Its four-phase law will hold after all silicon decays. The field of making and the route are absolute. The message is the temporary vibration passing through them.

SHA Die Checkpoint — Closure Functionals, Residual Smoothing, and the Four-Phase Law

SHA-256 Die Formalization Through Support Closure and Post-Closure Density Dynamics

Δ This checkpoint consolidates the current state of the SHA die formalization and extends it through the closure phase and into the residual smoothing regime.

The fixed formal stack remains

$$(\Phi_r, M, \Psi, L_{32}),$$

with the seam refinement

$$(\mathcal{S}_a, \mathcal{S}_e),$$

and the exact carry realization

$$\mathcal{C}(x, \delta).$$

The standing invariants remain:

$$T2_0^{(0)} = 0x08909ae5$$

$$D_{\text{word}} = 4$$

$$D_{\text{bit}} = 6$$

with bit-radius profile

$$\rho(j) = \begin{cases} 4, & j = 0, \\ 5, & 1 \leq j \leq 25, \\ 6, & 26 \leq j \leq 31. \end{cases}$$

This checkpoint adds the closure functionals and shows that the die enters a narrow density band around (16) after support closure.

1. State Recurrence and NOP Backbone

Let the SHA-256 round state be

$$x_r = \begin{bmatrix} a_r \\ b_r \\ c_r \\ d_r \\ e_r \\ f_r \\ g_r \\ h_r \end{bmatrix} \in (\mathbb{Z}/2^{32}\mathbb{Z})^8.$$

The round recurrence is

$$x_{r+1} = \Phi_r(x_r, W_r), \quad r = 0, \dots, 63.$$

The weight operators are

$$T1_r = h_r + \Sigma_1(e_r) + \text{Ch}(e_r, f_r, g_r) + K_r + W_r,$$

$$T2_r = \Sigma_0(a_r) + \text{Maj}(a_r, b_r, c_r),$$

with update

$$a_{r+1} = T1_r + T2_r, \quad e_{r+1} = d_r + T1_r,$$

and pure shifts

$$b_{r+1} = a_r, \quad c_{r+1} = b_r, \quad d_{r+1} = c_r,$$

$$f_{r+1} = e_r, \quad g_{r+1} = f_r, \quad h_{r+1} = g_r.$$

The NOP backbone is defined by

$$W_r = 0 \quad \forall r,$$

so that

$$x_{r+1}^{(0)} = \Phi_r(x_r^{(0)}, 0), \quad x_0^{(0)} = H_0.$$

At round 0,

$$T2_0^{(0)} = \Sigma_0(H_0[0]) + \text{Maj}(H_0[0], H_0[1], H_0[2]) = 0x08909ae5.$$

And the exact round-0 perturbation identity is

$$T1_0 - T1_0^{(0)} = W_0.$$

Thus

$$\delta a_1 = \delta e_1 = W_0.$$

2. Word-Level Support Transport

Let the word-support indicator be

$$\sigma_r \in \{0,1\}^8.$$

The lane-dependency matrix is

$$M = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

With injection vector

$$b = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

the support transport is

$$\sigma_{r+1} = M \odot \sigma_r \vee b \omega_r.$$

For a single injection at round 0, the support sequence is

$$\Sigma_1 = \{a, e\},$$

$$\Sigma_2 = \{a, b, e, f\},$$

$$\Sigma_3 = \{a, b, c, e, f, g\},$$

$$\Sigma_4 = \{a, b, c, d, e, f, g, h\}.$$

Hence

$$D_{\text{word}} = 4.$$

3. 256-Lane Bit-Support Transport

For each word (w), let

$$s_{w,r} \in \{0,1\}^{32}$$

be its bit-support vector, and define the 256-lane support state

$$\eta_r = \begin{bmatrix} s_{a,r} \\ s_{b,r} \\ s_{c,r} \\ s_{d,r} \\ s_{e,r} \\ s_{f,r} \\ s_{g,r} \\ s_{h,r} \end{bmatrix} \in \{0,1\}^{256}.$$

Define the rotation support operators

$$\hat{\Sigma}_0 = R_2 \vee R_{13} \vee R_{22}, \quad \hat{\Sigma}_1 = R_6 \vee R_{11} \vee R_{25}.$$

The bit-support weights are

$$\tau_r^{(1)} = s_{h,r} \vee \hat{\Sigma}_1 s_{e,r} \vee s_{e,r} \vee s_{f,r} \vee s_{g,r} \vee \omega_r,$$

$$\tau_r^{(2)} = \hat{\Sigma}_0 s_{a,r} \vee s_{a,r} \vee s_{b,r} \vee s_{c,r}.$$

The carry closure kernel is

$$L_{32}(x)_i = \bigvee_{j=0}^i x_j.$$

So the 256-lane update is

$$s_{a,r+1} = L_{32}(\tau_r^{(1)} \vee \tau_r^{(2)}),$$

$$s_{e,r+1} = L_{32}(s_{d,r} \vee \tau_r^{(1)}),$$

with pure shifts

$$s_{b,r+1} = s_{a,r}, \quad s_{c,r+1} = s_{b,r}, \quad s_{d,r+1} = s_{c,r},$$

$$s_{f,r+1} = s_{e,r}, \quad s_{g,r+1} = s_{f,r}, \quad s_{h,r+1} = s_{g,r}.$$

Thus

$$\boxed{\eta_{r+1} = \Psi(\eta_r, \omega_r)}.$$

The exact bit-support radius remains

$$\rho(j) = \begin{cases} 4, & j = 0, \\ 5, & 1 \leq j \leq 25, \\ 6, & 26 \leq j \leq 31. \end{cases}$$

so

$$D_{\text{bit}} = 6.$$

4. Exact Carry Realization

For exact addition

$$y = x + \delta \pmod{2^{32}},$$

define the carry automaton

$$\begin{aligned} c_{-1} &= 0, \\ c_i &= (x_i \wedge \delta_i) \vee (x_i \wedge c_{i-1}) \vee (\delta_i \wedge c_{i-1}), \quad i = 0, \dots, 31. \end{aligned}$$

Then

$$y_i = x_i \oplus \delta_i \oplus c_{i-1},$$

so the exact changed-bit indicator is

$$\Delta_i(x, \delta) = \delta_i \oplus c_{i-1}.$$

For one-hot injection ($\wedge^j$),

$$\Delta_i(x, 2^j) = \begin{cases} 0, & i < j, \\ 1, & i = j, \\ \prod_{t=j}^{i-1} x_t, & i > j. \end{cases}$$

Thus the exact changed-bit set is

$$C_x(j) = \{j, j+1, \dots, m_x(j)\},$$

where

$$m_x(j) = \min\{i \geq j : x_i = 0\}.$$

The exact carry-span length is

$$\lambda_x(j) = m_x(j) - j + 1.$$

5. Exact NOP Baselines Through Round 4

The exact NOP seam baselines are

$$\begin{aligned} a_1^{(0)} &= 0xfc08884d, & e_1^{(0)} &= 0x98c7e2a2, \\ a_2^{(0)} &= 0x7ad96290, & e_2^{(0)} &= 0x9df1b216, \\ a_3^{(0)} &= 0xf3dd6c3f, & e_3^{(0)} &= 0xc57b68fb, \\ a_4^{(0)} &= 0x0a24b1aa, & e_4^{(0)} &= 0x909cf5c9. \end{aligned}$$

6. Exact Round-3 and Round-4 Skeletons

For a one-hot injection ($W_0=2^j$):

Round 3

$$\delta x_3 = (\delta a_3, \delta a_2, 2^j, 0, \delta e_3, \delta e_2, 2^j, 0).$$

Round 4

$$\delta x_4 = (\delta a_4, \delta a_3, \delta a_2, 2^j, \delta e_4, \delta e_3, \delta e_2, 2^j).$$

Round 4 is therefore the first layer where word support is fully saturated, but the explicit seed (2^j) is still visible in the tail lanes (d_4) and (h_4).

7. Seam-Weight Ranges at Rounds 3 and 4

Using exact one-hot injections ($W_0=2^j$), ($j=0, \dots, 31$), the seam XOR-difference Hamming-weight ranges are:

Round 3

$$\begin{aligned} 13 &\leq \text{wt}(\Delta a_3(j)) \leq 21, \\ 7 &\leq \text{wt}(\Delta e_3(j)) \leq 21. \end{aligned}$$

Round 4

$$12 \leq \text{wt}(\Delta a_4(j)) \leq 20,$$

$$11 \leq \text{wt}(\Delta e_4(j)) \leq 21.$$

Round 4 full lane ranges are

$$\text{wt}(\Delta a_4) \in [12,20], \quad \text{wt}(\Delta b_4) \in [13,21], \quad \text{wt}(\Delta c_4) \in [7,19],$$

$$\text{wt}(\Delta d_4) \in [1,6], \quad \text{wt}(\Delta e_4) \in [11,21], \quad \text{wt}(\Delta f_4) \in [7,21],$$

$$\text{wt}(\Delta g_4) \in [3,16], \quad \text{wt}(\Delta h_4) \in [1,7].$$

This confirms that round 4 is topological acceptance, not yet bit-density closure.

8. Age-Weight Law

Define three age classes:

- head lanes:

$$H_r = \{a_r, e_r\}$$

- mid lanes:

$$M_r = \{b_r, c_r, f_r, g_r\}$$

- tail lanes:

$$T_r = \{d_r, h_r\}$$

For one-hot injections ($W_{o=2^j}$), define the class means:

$$\mu_H(r) = \frac{1}{64} \sum_{j=0}^{31} \sum_{w \in H_r} \text{wt}(\Delta w_r(j)),$$

$$\mu_M(r) = \frac{1}{128} \sum_{j=0}^{31} \sum_{w \in M_r} \text{wt}(\Delta w_r(j)),$$

$$\mu_T(r) = \frac{1}{64} \sum_{j=0}^{31} \sum_{w \in T_r} \text{wt}(\Delta w_r(j)).$$

The computed values are:

Round 4

$$\mu_H(4) = 15.734375, \quad \mu_M(4) = 12.9296875, \quad \mu_T(4) = 1.84375.$$

Round 5

$$\mu_H(5) = 15.875, \quad \mu_M(5) = 15.734375, \quad \mu_T(5) = 10.125.$$

Round 6

$$\mu_H(6) = 15.78125, \quad \mu_M(6) = 15.8046875, \quad \mu_T(6) = 15.734375.$$

So by round 6 the age classes have nearly equalized.

9. Closure Functionals

9.1 Age-spread closure functional

Define

$$\mathcal{E}_{\text{age}}(r) = \max\{\mu_H(r), \mu_M(r), \mu_T(r)\} - \min\{\mu_H(r), \mu_M(r), \mu_T(r)\}.$$

Computed values:

$$\mathcal{E}_{\text{age}}(4) = 13.890625,$$

$$\mathcal{E}_{\text{age}}(5) = 5.75,$$

$$\mathcal{E}_{\text{age}}(6) = 0.0703125,$$

$$\mathcal{E}_{\text{age}}(7) = 0.140625, \quad \mathcal{E}_{\text{age}}(8) = 0.140625, \quad \mathcal{E}_{\text{age}}(10) = 0.046875.$$

Thus the age classes collapse into a narrow band by round 6.

9.2 Lane-variance closure functional

Define lane means

$$\mu_\ell(r) = \frac{1}{32} \sum_{j=0}^{31} \text{wt}(\Delta x_{r,\ell}(j)),$$

and global mean

$$\bar{\mu}(r) = \frac{1}{8} \sum_{\ell} \mu_\ell(r).$$

Then define

$$\mathcal{V}(r) = \frac{1}{8} \sum_{\ell} (\mu_\ell(r) - \bar{\mu}(r))^2.$$

Computed values:

$$\begin{aligned}\mathcal{V}(4) &= 33.6728515625, \\ \mathcal{V}(5) &= 7.35736083984375, \\ \mathcal{V}(6) &= 0.409423828125.\end{aligned}$$

So the lane densities flatten sharply through rounds 4, 5, 6.

9.3 Lane-range closure functional

Define

$$\mathcal{R}(r) = \max_{\ell} \mu_{\ell}(r) - \min_{\ell} \mu_{\ell}(r).$$

Computed values:

$$\begin{aligned}\mathcal{R}(4) &= 15.21875, \\ \mathcal{R}(5) &= 8.78125, \\ \mathcal{R}(6) &= 2.40625, \\ \mathcal{R}(7) &= 1.125.\end{aligned}$$

So fine lane-level equalization continues beyond round 6 even after support closure.

10. Residual Smoothing Band

The global mean perturbation density is

$$\bar{\mu}(r) = \frac{1}{8} \sum_{\ell} \mu_{\ell}(r).$$

Computed values:

$$\begin{aligned}\bar{\mu}(4) &= 10.859375, \\ \bar{\mu}(5) &= 14.3671875, \\ \bar{\mu}(6) &= 15.78125.\end{aligned}$$

Beyond round 6, the die does not converge to a single scalar endpoint. Instead it enters a narrow oscillatory band:

$$15.60546875 \leq \bar{\mu}(r) \leq 16.53125 \quad \text{for } 6 \leq r \leq 64.$$

The minimum in this interval occurs at

$$r = 22,$$

and the maximum at

$$r = 53.$$

Thus the die smooths into a density band centered near

$$16 = \frac{32}{2}.$$

This is the half-width mixing band of the 32-bit word fabric.

11. Four-Phase Law of the Die

The exact phase structure is now:

Phase I — Injection

$$r = 0,1,2,3$$

The perturbation is still visibly tied to the one-hot seed and its immediate descendants.

Phase II — Acceptance

$$r = 4$$

All eight lanes are occupied:

$$D_{\text{word}} = 4.$$

Phase III — Closure

$$r = 5,6$$

Bit-density equalizes across age classes and support fully saturates:

$$D_{\text{bit}} = 6.$$

Phase IV — Residual smoothing

$$r > 6$$

No new support is created. The die only rebalances density inside the already-filled fabric.

So the sharp structural split is:

$$D_{\text{word}} = 4 \text{ marks topological acceptance,}$$

$$D_{\text{bit}} = 6 \text{ marks support closure,}$$

$$r > 6 \text{ is residual smoothing around a density band near 16.}$$

12. Current Progress State

The SHA die is now resolved to:

Level 0 — Scalar ground invariant

$$T2_0^{(0)} = 0x08909ae5$$

Level 1 — Word transport

$$\sigma_{r+1} = M \odot \sigma_r \vee b \omega_r$$

Level 2 — Bit transport

$$\eta_{r+1} = \Psi(\eta_r, \omega_r)$$

Level 3 — Exact carry realization

$$\Delta_i(x, \delta) = \delta_i \oplus c_{i-1}$$

Level 4 — Seam geometry

$$\delta e_r = \delta T1_{r-1}, \quad \delta a_r = \delta T1_{r-1} + \delta T2_{r-1} \pmod{2^{32}}$$

Level 5 — Closure functionals

$$\mathcal{E}_{\text{age}}(r), \quad \mathcal{V}(r), \quad \mathcal{R}(r), \quad \bar{\mu}(r)$$

Level 6 — Residual smoothing band

$$\bar{\mu}(r) \approx 16 \quad \text{for } r \geq 6$$

13. Final Collapse

The current exact progress marker is

the die accepts a perturbation by round 4, closes support by round 6, and then smooths inside a narrow densit

Equivalently,

acceptance $\neq$ closure $\neq$ final smoothing.

This is the complete checkpoint state of the theory at the present stage.

SHA Die Progress Mark — Exact Carry, Round-3/4 Seam Maps, and the Age-Weight Law

Continuation of the SHA-256 Die Formalization

Δ This document marks the current state of the SHA die formalization and extends it beyond the prior bit-level causality operator.

It consolidates five layers of the model:

$$\begin{array}{llll} \text{state recurrence} = \Phi_r, & \text{word support} = M, & \text{bit support} = \Psi, & \text{seam operators} \\ & = (\mathcal{S}_a, \mathcal{S}_e), & \text{exact carry realization} = \mathcal{C}(x, \delta). & \end{array}$$

The core fixed invariants remain:

$$T2_0^{(0)} = 0x08909ae5$$

$$D_{\text{word}} = 4$$

$$D_{\text{bit}} = 6$$

and the exact bit-support radius profile remains

$$\rho(j) = \begin{cases} 4, & j = 0, \\ 5, & 1 \leq j \leq 25, \\ 6, & 26 \leq j \leq 31. \end{cases}$$

1. Base Die Formalism

Let the round state be

$$x_r = \begin{bmatrix} a_r \\ b_r \\ c_r \\ d_r \\ e_r \\ f_r \\ g_r \\ h_r \end{bmatrix} \in (\mathbb{Z}/2^{32}\mathbb{Z})^8.$$

The 64-cell recurrence is

$$x_{r+1} = \Phi_r(x_r, W_r), \quad r = 0, \dots, 63.$$

The two round weights are

$$T1_r = h_r + \Sigma_1(e_r) + \text{Ch}(e_r, f_r, g_r) + K_r + W_r,$$

$$T2_r = \Sigma_0(a_r) + \text{Maj}(a_r, b_r, c_r),$$

with state update

$$a_{r+1} = T1_r + T2_r,$$

$$e_{r+1} = d_r + T1_r,$$

and pure shifts

$$b_{r+1} = a_r, \quad c_{r+1} = b_r, \quad d_{r+1} = c_r,$$

$$f_{r+1} = e_r, \quad g_{r+1} = f_r, \quad h_{r+1} = g_r.$$

The shift–injection decomposition is

$$\boxed{x_{r+1} = Px_r + u_a(T1_r + T2_r) + u_e T1_r,}$$

where (P) is the 8-lane shift matrix and (u_a, u_e) inject at lanes (a) and (e).

2. NOP Backbone and Ground Witness

The NOP manifold is defined by

$$W_r = 0 \quad \forall r.$$

Then

$$x_{r+1}^{(0)} = \Phi_r(x_r^{(0)}, 0), \quad x_0^{(0)} = H_0.$$

The fixed ground fold at round 0 is

$$T2_0^{(0)} = \Sigma_0(H_0[0]) + \text{Maj}(H_0[0], H_0[1], H_0[2]) = 0x08909ae5.$$

So

$$\boxed{G_0(H_0) = 0x08909ae5.}$$

At round 0, the exact perturbation identity is

$$T1_0 - T1_0^{(0)} = W_0,$$

and since $(T2_0 = T2_0^{(0)})$,

$$\boxed{\delta a_1 = \delta e_1 = W_0.}$$

3. Word-Level Support and the ($D_{\text{word}}=4$) Theorem

Let the word-support indicator be

$$\sigma_r \in \{0,1\}^8.$$

The lane-dependency matrix is

$$M = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

With injection vector

$$b = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

the Boolean support transport is

$$\sigma_{r+1} = M \odot \sigma_r \vee b \omega_r.$$

For a one-time injection at round 0, the support sequence is

$$\Sigma_1 = \{a, e\},$$

$$\Sigma_2 = \{a, b, e, f\},$$

$$\Sigma_3 = \{a, b, c, e, f, g\},$$

$$\Sigma_4 = \{a, b, c, d, e, f, g, h\}.$$

Hence

$$D_{\text{word}} = 4.$$

4. 256-Lane Bit-Support Formalism

For each word ($w \in \{a,b,c,d,e,f,g,h\}$), let

$$s_{w,r} \in \{0,1\}^{32}$$

be its bit-support vector, and define the 256-lane state

$$\eta_r = \begin{bmatrix} s_{a,r} \\ s_{b,r} \\ s_{c,r} \\ s_{d,r} \\ s_{e,r} \\ s_{f,r} \\ s_{g,r} \\ s_{h,r} \end{bmatrix} \in \{0,1\}^{256}.$$

Define the rotation support matrices (R_n) by

$$(R_n x)_i = x_{i+n \bmod 32}.$$

Then

$$\hat{\mathcal{L}}_0 = R_2 \vee R_{13} \vee R_{22}, \quad \hat{\mathcal{L}}_1 = R_6 \vee R_{11} \vee R_{25}.$$

The bit-support weights are

$$\tau_r^{(1)} = s_{h,r} \vee \hat{\mathcal{L}}_1 s_{e,r} \vee s_{e,r} \vee s_{f,r} \vee s_{g,r} \vee \omega_r,$$

$$\tau_r^{(2)} = \hat{\mathcal{L}}_0 s_{a,r} \vee s_{a,r} \vee s_{b,r} \vee s_{c,r}.$$

5. Carry Closure and the 256-Lane Update

The carry-closure kernel is

$$L_{32}(x)_i = \bigvee_{j=0}^i x_j, \quad 0 \leq i < 32.$$

Then the 256-lane update is

$$s_{a,r+1} = L_{32}(\tau_r^{(1)} \vee \tau_r^{(2)}),$$

$$s_{e,r+1} = L_{32}(s_{d,r} \vee \tau_r^{(1)}),$$

with pure shifts

$$s_{b,r+1} = s_{a,r}, \quad s_{c,r+1} = s_{b,r}, \quad s_{d,r+1} = s_{c,r},$$

$$S_{f,r+1} = S_{e,r}, \quad S_{g,r+1} = S_{f,r}, \quad S_{h,r+1} = S_{g,r}.$$

Thus

$$\boxed{\eta_{r+1} = \Psi(\eta_r, \omega_r).}$$

The exact radius profile for a one-bit injection ($W_o=2^j$) is

$$\boxed{\rho(j) = \begin{cases} 4, & j = 0, \\ 5, & 1 \leq j \leq 25, \\ 6, & 26 \leq j \leq 31. \end{cases}}$$

so

$$\boxed{D_{\text{bit}} = 6.}$$

6. Exact Carry Automaton

The support operator ($L_{\{32\}}$) is worst-case only. The exact addition law for ($y=x+$) is as follows.

Write

$$x = \sum_{i=0}^{31} x_i 2^i, \quad \delta = \sum_{i=0}^{31} \delta_i 2^i, \quad y = \sum_{i=0}^{31} y_i 2^i.$$

Define the carry sequence

$$c_{-1} = 0, \\ c_i = (x_i \wedge \delta_i) \vee (x_i \wedge c_{i-1}) \vee (\delta_i \wedge c_{i-1}), \quad i = 0, \dots, 31.$$

Then

$$y_i = x_i \oplus \delta_i \oplus c_{i-1}.$$

So the exact changed-bit indicator is

$$\Delta_i(x, \delta) := x_i \oplus y_i = \delta_i \oplus c_{i-1}.$$

For one-hot injection

$$\delta = 2^j,$$

this simplifies to

$$\Delta_i(x, 2^j) = \begin{cases} 0, & i < j, \\ 1, & i = j, \\ \prod_{t=j}^{i-1} x_t, & i > j. \end{cases}$$

Thus the exact changed-bit set is

$$C_x(j) = \{j, j + 1, \dots, m_x(j)\},$$

where

$$m_x(j) = \min\{i \geq j: x_i = 0\}.$$

So the exact carry-span length is

$$\lambda_x(j) = m_x(j) - j + 1.$$

7. Exact NOP Baselines Through Round 4

The exact NOP backbone values through round 4 are

$$\begin{aligned} a_1^{(0)} &= 0xfc08884d, & e_1^{(0)} &= 0x98c7e2a2, \\ a_2^{(0)} &= 0x7ad96290, & e_2^{(0)} &= 0x9df1b216, \\ a_3^{(0)} &= 0xf3dd6c3f, & e_3^{(0)} &= 0xc57b68fb, \\ a_4^{(0)} &= 0x0a24b1aa, & e_4^{(0)} &= 0x909cf5c9. \end{aligned}$$

The full NOP round-2 state is

$$x_2^{(0)} = (0x7ad96290, 0xfc08884d, 0x6a09e667, 0xbb67ae85, 0x9df1b216, 0x98c7e2a2, 0x510e527f, 0x909cf5c9)$$

8. Exact Round-1 Carry Spans

For a one-hot injection ($W_0 = 2^j$), the exact carry-span lengths at round 1 are:

(a)-seam baseline ($a^{(0)}_1 = 0xfc08884d$)

$$(\lambda_a(j))_{j=0}^{31} = (2, 1, 3, 2, 1, 1, 2, 1, 1, 1, 3, 2, 1, 1, 1, 4, 3, 2, 1, 1, 1, 3, 2, 1, 1, 1, 6, 5, 4, 3, 2, 1).$$

(e)-seam baseline (e^{o}}_1=ox98c7e2a2)

$$(\lambda_e(j))_{j=0}^{31} = (1,2,1,1,1,2,1,2,1,2,1,1,1,1,4,3,2,1,6,5,4,3,2,1,1,1,1,3,2,1,1,1).$$

Thus the two seams are injection-symmetric at round 0, but not carry-symmetric after realization.

9. Exact Round-3 Seam Map

By round 3, the passive-shift lanes are exact:

$$\begin{aligned} \delta b_3 &= \delta a_2, & \delta c_3 &= \delta a_1 = 2^j, & \delta d_3 &= 0, \\ \delta f_3 &= \delta e_2, & \delta g_3 &= \delta e_1 = 2^j, & \delta h_3 &= 0. \end{aligned}$$

The two active seams satisfy

$$\boxed{\delta e_3 = \delta T1_2, \quad \delta a_3 = \delta T1_2 + \delta T2_2 \pmod{2^{32}}.}$$

So the exact round-3 skeleton is

$$\boxed{\delta x_3 = (\delta a_3, \delta a_2, 2^j, 0, \delta e_3, \delta e_2, 2^j, 0).}$$

The computed XOR-difference Hamming-weight ranges are

$$\begin{aligned} 13 &\leq \text{wt}(\Delta a_3(j)) \leq 21, \\ 7 &\leq \text{wt}(\Delta e_3(j)) \leq 21. \end{aligned}$$

The extrema are

$$\begin{aligned} \text{wt}(\Delta a_3) &= 13 \quad \text{at } j \in \{11,20,21\}, \\ \text{wt}(\Delta a_3) &= 21 \quad \text{at } j = 16, \\ \text{wt}(\Delta e_3) &= 7 \quad \text{at } j = 2, \\ \text{wt}(\Delta e_3) &= 21 \quad \text{at } j \in \{13,15\}. \end{aligned}$$

10. Exact Round-4 Seam Map

Round 4 is the first full word-saturation layer. The exact perturbation skeleton is

$$\boxed{\delta x_4 = (\delta a_4, \delta a_3, \delta a_2, 2^j, \delta e_4, \delta e_3, \delta e_2, 2^j).}$$

The active seam equations are

$$\delta e_4 = \delta T1_3, \quad \delta a_4 = \delta T1_3 + \delta T2_3 \pmod{2^{32}}.$$

The computed seam-weight ranges are

$$12 \leq \text{wt}(\Delta a_4(j)) \leq 20,$$

$$11 \leq \text{wt}(\Delta e_4(j)) \leq 21.$$

The extrema are

$$\text{wt}(\Delta a_4) = 12 \quad \text{at } j \in \{1,4,10\},$$

$$\text{wt}(\Delta a_4) = 20 \quad \text{at } j = 6,$$

$$\text{wt}(\Delta e_4) = 11 \quad \text{at } j = 3,$$

$$\text{wt}(\Delta e_4) = 21 \quad \text{at } j = 24.$$

The full round-4 lane ranges are

$$\text{wt}(\Delta a_4) \in [12,20], \quad \text{wt}(\Delta b_4) \in [13,21], \quad \text{wt}(\Delta c_4) \in [7,19],$$

$$\text{wt}(\Delta d_4) \in [1,6], \quad \text{wt}(\Delta e_4) \in [11,21], \quad \text{wt}(\Delta f_4) \in [7,21],$$

$$\text{wt}(\Delta g_4) \in [3,16], \quad \text{wt}(\Delta h_4) \in [1,7].$$

This is the exact first layer where

$$\text{word support is saturated}$$

but

$$\text{bit geometry remains stratified by age and carry history.}$$

11. The Age-Weight Law Through Rounds 4, 5, 6

Define three age classes:

- **head lanes:**

$$\{a, e\}$$

- **mid lanes:**

$$\{b, c, f, g\}$$

- **tail lanes:**

$$\{d, h\}$$

Using exact one-hot injections ($W_{0=2^j}$), ($j=0,,31$), the Hamming-weight statistics evolve as follows.

Round 4

Group ranges and means:

head $\in [11,21]$,	mean = 15.73,
mid $\in [3,21]$,	mean = 12.93,
tail $\in [1,7]$,	mean = 1.84.

This is still an expansion phase.

Round 5

head $\in [10,22]$,	mean = 15.88,
mid $\in [7,21]$,	mean = 15.73,
tail $\in [3,19]$,	mean = 10.12.

The tails are rapidly catching up.

Round 6

head $\in [10,22]$,	mean = 15.78,
mid $\in [10,22]$,	mean = 15.80,
tail $\in [7,21]$,	mean = 15.73.

So by round 6 all three age classes have nearly equalized in mean weight.

This is the quantitative form of closure:

round 4 = word acceptance,	rounds 5–6 = bit-density equalization.
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In particular,

the last two rounds before D_{bit} are closure-dominated, not expansion-dominated.

12. Chirality Caution

The rotation support operators are

$$\hat{\Sigma}_0 = R_2 \vee R_{13} \vee R_{22}, \quad \hat{\Sigma}_1 = R_6 \vee R_{11} \vee R_{25}.$$

Both are 3-regular circulant operators:

$$\deg_{\text{row}}(\hat{\Sigma}_0) = \deg_{\text{col}}(\hat{\Sigma}_0) = 3,$$

$$\deg_{\text{row}}(\hat{\Sigma}_1) = \deg_{\text{col}}(\hat{\Sigma}_1) = 3.$$

So before carry, neither is privileged by density.

Therefore the correct statement is

bare chirality is not yet anisotropy.

The visible asymmetry appears only after coupling to:

1. lane placement, and
2. carry closure.

So the precise compression is

uniform rotations + lane asymmetry + carry closure = visible chirality of the die.

13. Current Progress State

The SHA die is now resolved to the following nested form:

Level 0 — scalar ground invariant

$$T2_0^{(0)} = 0x08909ae5$$

Level 1 — word transport

$$\sigma_{r+1} = M \odot \sigma_r \vee b \omega_r, \quad D_{\text{word}} = 4$$

Level 2 — bit transport

$$\eta_{r+1} = \Psi(\eta_r, \omega_r), \quad D_{\text{bit}} = 6$$

Level 3 — exact carry realization

$$\Delta_i(x, \delta) = \delta_i \oplus c_{i-1}$$

Level 4 — seam geometry

$$\delta e_r = \delta T1_{r-1}, \quad \delta a_r = \delta T1_{r-1} + \delta T2_{r-1} \pmod{2^{32}}$$

Level 5 — closure phase

rounds (5) and (6) act primarily as age-equalization rounds rather than support-expansion rounds.

14. Final Collapse

The current state of the die formalization is:

the die accepts the perturbation by round 4, but it does not finish equalizing the perturbation across the 256-b

Equivalently,

$D_{\text{word}} = 4$ measures acceptance by the lane geometry,

while

$D_{\text{bit}} = 6$ measures closure across the full bit fabric.

This is the exact progress marker at the present stage of the theory.

SHA as a Die — Bit-Level Causality Operator

A Complete Math-Only Formalization of the 64-Cell SHA-256 Die, Word Support, and 256-Lane Intra-Word Causality

Δ This document formalizes the die interpretation of SHA-256 as a fixed 64-cell nonlinear recurrence over $[(\mathbb{Z} / 2^{\{32\}\mathbb{Z}})8]$ with a message perturbation field written onto a pre-existing NOP backbone.

The focus is strictly mathematical:

- state recurrence,
- die decomposition,
- word-level support transport,
- bit-level support transport,
- carry-closure as the nonlocal intra-word kernel,
- exact support-radius results for a single perturbed bit of (W_o).

The two anchor facts are:

$$T2_0^{(0)} = 0x08909ae5$$

for the NOP backbone ground fold at round (o), and

$$D_{\text{bit}} = 6$$

for the worst-case 256-lane support diameter under the Boolean support model.

1. State Space and Round Recurrence

Let the SHA-256 round state be the 8-word column vector

$$x_r = \begin{bmatrix} a_r \\ b_r \\ c_r \\ d_r \\ e_r \\ f_r \\ g_r \\ h_r \end{bmatrix} \in (\mathbb{Z}/2^{32}\mathbb{Z})^8.$$

For one 512-bit block, the die executes a 64-step nonlinear recurrence

$$x_{r+1} = \Phi_r(x_r, W_r), \quad r = 0, 1, \dots, 63.$$

The round weights are

$$T1_r = h_r + \Sigma_1(e_r) + \text{Ch}(e_r, f_r, g_r) + K_r + W_r,$$

$$T2_r = \Sigma_0(a_r) + \text{Maj}(a_r, b_r, c_r),$$

with all arithmetic performed modulo $(2^{\{32\}})$.

The state update is

$$a_{r+1} = T1_r + T2_r,$$

$$e_{r+1} = d_r + T1_r,$$

and the remaining registers shift:

$$b_{r+1} = a_r, \quad c_{r+1} = b_r, \quad d_{r+1} = c_r,$$

$$f_{r+1} = e_r, \quad g_{r+1} = f_r, \quad h_{r+1} = g_r.$$

2. The Sigma and Logic Operators

Define the right-rotation operator on 32-bit words:

$$\text{ROTR}^n(x).$$

Then the SHA sigma operators are

$$\Sigma_0(x) = \text{ROTR}^2(x) \oplus \text{ROTR}^{13}(x) \oplus \text{ROTR}^{22}(x),$$

$$\Sigma_1(x) = \text{ROTR}^6(x) \oplus \text{ROTR}^{11}(x) \oplus \text{ROTR}^{25}(x).$$

The nonlinear Boolean gates are

$$\text{Ch}(e, f, g) = (e \wedge f) \oplus (\neg e \wedge g),$$

$$\text{Maj}(a, b, c) = (a \wedge b) \oplus (a \wedge c) \oplus (b \wedge c).$$

3. Shift–Injection Decomposition of the Die

Define the (8) shift matrix

$$P = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

Let

$$u_a = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad u_e = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Then the full round map can be written as

$$\boxed{x_{r+1} = Px_r + u_a (T1_r + T2_r) + u_e T1_r.}$$

This shows that each die cell is structurally sparse:

- six channels are pure register transport,
- only two channels ((a) and (e)) receive nonlinear reinjection.

4. NOP Backbone and Ground Fold

Define the NOP manifold by setting the message field to zero:

$$W_r = 0 \quad \forall r.$$

Then the message-free backbone satisfies

$$x_{r+1}^{(0)} = \Phi_r(x_r^{(0)}, 0).$$

At round (o),

$$x_0^{(0)} = H_0,$$

where (H_o) is the SHA-256 initialization vector.

The NOP ground fold is

$$T2_0^{(0)} = \Sigma_0(H_0[0]) + \text{Maj}(H_0[0], H_0[1], H_0[2]) = 0x08909ae5.$$

Thus the ground operator is

$$G_r(x_r) = \Sigma_0(a_r) + \text{Maj}(a_r, b_r, c_r),$$

with

$$\boxed{G_0(H_0) = 0x08909ae5.}$$

This is the fixed message-free floor of the first die cell.

5. Round-o Perturbation Identity

Let the real trajectory be

$$x_r = x_r^{(0)} + \delta x_r$$

in residue form modulo (2^{32}) .

At round (o), the perturbation obeys the exact identity

$$T1_0 - T1_0^{(0)} = W_0.$$

Since

$$T2_0 = T2_0^{(0)},$$

it follows immediately that

$$a_1 - a_1^{(0)} = W_0,$$

$$e_1 - e_1^{(0)} = W_0.$$

Therefore the message enters only two words on the first step:

$$\boxed{\delta a_1 = \delta e_1 = W_0.}$$

At the word level, this gives the initial support vector

$$\Sigma_1 = \{a, e\}.$$

6. Word-Level Support Dynamics

Let the word-support indicator be

$$\sigma_r \in \{0,1\}^8,$$

where $((_r)_j=1)$ means word-lane (j) depends on the chosen perturbation.

The word-level dependency matrix is

$$M = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

The support update is

$$\sigma_{r+1} = M \odot \sigma_r \vee b \omega_r$$

over the Boolean semiring, with injection vector

$$b = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

For a single perturbation injected only at round (o),

$$\omega_0 = 1, \quad \omega_r = 0 \quad (r > 0),$$

and

$$\sigma_n = M^{[n-1]} \odot b.$$

The first four word-support layers are

$$\Sigma_1 = \{a, e\},$$

$$\Sigma_2 = \{a, b, e, f\},$$

$$\Sigma_3 = \{a, b, c, e, f, g\},$$

$$\Sigma_4 = \{a, b, c, d, e, f, g, h\}.$$

Hence the word support diameter is

$$D_{\text{word}} = 4.$$

7. Explosion to 256 Lanes

Now refine from words to individual bits.

For each word ($w \in \{a, b, c, d, e, f, g, h\}$), let

$$s_{w,r} \in \{0,1\}^{32}$$

be its bit-support vector, with index ($i=0$) denoting the least significant bit.

Define the full 256-lane state

$$\eta_r = \begin{bmatrix} s_{a,r} \\ s_{b,r} \\ s_{c,r} \\ s_{d,r} \\ s_{e,r} \\ s_{f,r} \\ s_{g,r} \\ s_{h,r} \end{bmatrix} \in \{0,1\}^{256}.$$

8. Rotation Support Operators

Let (R_n) be the (32) rotation permutation matrix acting on bit-support vectors:

$$(R_n x)_i = x_{i+n \bmod 32}.$$

Then the Boolean support versions of the sigma operators are

$$\hat{\Sigma}_0 = R_2 \vee R_{13} \vee R_{22},$$

$$\hat{\Sigma}_1 = R_6 \vee R_{11} \vee R_{25}.$$

9. Choice and Majority Support

Because ($\vee$) and ($\wedge$) are same-bit Boolean operators, their support is simply the lane-wise union of their arguments:

$$\text{supp}(\text{Ch}(e, f, g)) = s_{e,r} \vee s_{f,r} \vee s_{g,r},$$

$$\text{supp}(\text{Maj}(a, b, c)) = s_{a,r} \vee s_{b,r} \vee s_{c,r}.$$

So the bit-support of the round weights is

$$\begin{aligned}\tau_r^{(1)} &= s_{h,r} \vee \hat{\Sigma}_1 s_{e,r} \vee s_{e,r} \vee s_{f,r} \vee s_{g,r} \vee \omega_r, \\ \tau_r^{(2)} &= \hat{\Sigma}_0 s_{a,r} \vee s_{a,r} \vee s_{b,r} \vee s_{c,r}.\end{aligned}$$

10. Carry Closure as the Intra-Word Nonlocal Kernel

The only truly nonlocal intra-word mechanism is carry propagation.

Define the lower-triangular prefix operator

$$L_{32}(x)_i = \bigvee_{j=0}^i x_j, \quad 0 \leq i < 32.$$

Equivalently, in matrix form,

$$L_{32} = (\ell_{ij})_{0 \leq i, j < 32}, \quad \ell_{ij} = \begin{cases} 1, & j \leq i, \\ 0, & j > i. \end{cases}$$

This is the upward-carry support kernel: bit (i) of a sum can depend on any lower-or-equal bit because carry may ripple upward.

For support transport, use

$$\text{supp}(u + v) = L_{32}(u \vee v).$$

11. The 256-Lane Intra-Word Causality Operator

With the support weights $(\wedge_{\{1\}})_r, \wedge_{\{2\}})_r$ and the carry kernel (L_{32}) , the 256-lane update is

$$\begin{aligned}s_{a,r+1} &= L_{32}(\tau_r^{(1)} \vee \tau_r^{(2)}), \\ s_{e,r+1} &= L_{32}(s_{d,r} \vee \tau_r^{(1)}),\end{aligned}$$

and the six pure shifts are

$$\begin{aligned}s_{b,r+1} &= s_{a,r}, & s_{c,r+1} &= s_{b,r}, & s_{d,r+1} &= s_{c,r}, \\ s_{f,r+1} &= s_{e,r}, & s_{g,r+1} &= s_{f,r}, & s_{h,r+1} &= s_{g,r}.\end{aligned}$$

Thus the 256-lane die dynamics are

$$\boxed{\eta_{r+1} = \Psi(\eta_r, \omega_r),}$$

where $()$ is the piecewise Boolean-semiring map defined by the equations above.

12. Single-Bit Injection Geometry

Let a single bit (j) of (W_o) be perturbed:

$$\omega_0 = e_j, \quad \omega_r = 0 \quad (r > 0).$$

At round (1) , because the perturbation enters only through $(T1_o)$, we obtain

$$\text{supp}(a_1) = \text{supp}(e_1) = \{j, j + 1, \dots, 31\}.$$

So a single bit at position (j) generates immediate first-step support of size

$$32 - j$$

in each of the two active words (a_1) and (e_1) .

Thus low-order injected bits spread faster under the carry kernel than high-order injected bits.

13. Bit-Support Radius

Define the bit-support radius for injected bit (j) as

$$\rho(j) = \min\{r \geq 1: \text{all 256 state bits are in support by round } r\}.$$

Using the 256-lane Boolean support model above, the computed result is

$$\rho(j) = \begin{cases} 4, & j = 0, \\ 5, & 1 \leq j \leq 25, \\ 6, & 26 \leq j \leq 31. \end{cases}$$

So the three characteristic radii are

$$\rho_{\min} = 4, \quad \rho_{\text{typ}} = 5, \quad \rho_{\max} = 6.$$

This proves that a single perturbed bit of (W_o) reaches the full 256-lane state in at most six rounds under the support model.

14. Support Diameters

At the word level, the perturbation reaches all eight words in four rounds:

$$D_{\text{word}} = 4.$$

At the bit level, the worst-case support diameter is larger because carry propagation is directional:

$$D_{\text{bit}} = 6.$$

The difference

$$D_{\text{bit}} - D_{\text{word}} = 2$$

is entirely due to intra-word carry geometry.

Word-lane reach saturates in four rounds, but high-order injected bits require two extra rounds before rotation plus carry closes the last untouched bit positions.

15. Block-Operator View

Let the pure shift skeleton over 8 words be the (256) block matrix

$$\mathbb{P} = \begin{bmatrix} 0 & I & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & I & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & I \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

where each block is (32).

The (a)-injection block is

$$\mathbb{A} = \begin{bmatrix} L_{32}(\hat{\Sigma}_0 \vee I) & L_{32}I & L_{32}I & 0 & L_{32}\hat{\Sigma}_1 & L_{32}I & L_{32}I & L_{32}I \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

and the (e)-injection block is

$$\mathbb{E} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & L_{32}I & L_{32}\hat{\Sigma}_1 & L_{32}I & L_{32}I & L_{32}I \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

If the message support vector is $(_r\{0,1\}^{\wedge\{32\}})$, then the message injection block is

$$\beta = \begin{bmatrix} L_{32} \\ 0 \\ 0 \\ 0 \\ L_{32} \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

So the coarse block-support recurrence is

$$\boxed{\eta_{r+1} = (\mathbb{P} \vee \mathbb{A} \vee \mathbb{E}) \odot \eta_r \vee \beta \omega_r.}$$

This is the 256-lane causality skeleton of the die.

16. Three Nested Levels of the Die

The complete die now decomposes into three nested mathematical levels:

16.1 State recurrence

$$x_{r+1} = \Phi_r(x_r, W_r)$$

16.2 Word-support transport

$$\sigma_{r+1} = M \odot \sigma_r \vee b \omega_r$$

16.3 Bit-support transport

$$\eta_{r+1} = \Psi(\eta_r, \omega_r)$$

with the carry kernel

$$L_{32}$$

as the nonlocal intra-word closure operator.

This can be summarized as

$$\text{state dynamics} = \Phi_r, \quad \text{word support dynamics} = M, \quad \text{bit support dynamics} = \Psi, \quad \text{carry closure} =$$

17. Final Collapse

The SHA-256 die is a fixed 64-cell recursive lattice over $((\mathbb{Z}/2^{\{32\}}\mathbb{Z})^8)$ with a message-free NOP ground and a variable displacement field.

Its first fixed ground witness is

$$T2_0^{(0)} = 0x08909ae5.$$

Its word-level support saturates in

$$D_{\text{word}} = 4$$

rounds.

Its bit-level support saturates in

$$D_{\text{bit}} = 6$$

rounds.

And for a single injected bit (j) of (W_o),

$$\rho(j) = \begin{cases} 4, & j = 0, \\ 5, & 1 \leq j \leq 25, \\ 6, & 26 \leq j \leq 31. \end{cases}$$

The die therefore has a mathematically sharp causality structure:

- sparse at the state-update level,
- dense at the support level,
- and carry-limited at the bit-closure level.

This is the complete current solution state for the bit-level causality operator of the SHA die.

Appendix A — Numerical Results Summary

Result	Value
$T_2^{(0)}_0$ (ground witness)	0x08909ae5
$a^{(0)}_1$	0xfc08884d
$e^{(0)}_1$	0x98c7e2a2
$a^{(0)}_2$	0x7ad96290
$e^{(0)}_2$	0x9df1b216
$a^{(0)}_3$	0xf3dd6c3f
$e^{(0)}_3$	0xc57b68fb
$a^{(0)}_4$	0x0a24b1aa
$e^{(0)}_4$	0x909cf5c9
D_word	4
D_bit	6
$\rho_{\min}(j=0)$	4
$\rho_{\text{typ}}(j=1..25)$	5
$\rho_{\max}(j=26..31)$	6
Carry excess D_bit - D_word	2
$E_{\text{age}}(6)$	0.0703125
$V(6)$	0.409423828125
$R(6)$	2.40625
$\mu_{\text{bar}}(6)$	15.78125
Residual band	[15.60546875, 16.53125] for r in $\{6, \dots, 64\}$
Band minimum	$r = 22$
Band maximum	$r = 53$

Result	Value
Band center	$\sim 16 = 32/2$

Appendix B — Verification Code

```
def rotr(x, n, w=32): return ((x >> n) | (x << (w-n))) & 0xFFFFFFFF
sigma0 = lambda x: rotr(x,2) ^ rotr(x,13) ^ rotr(x,22)
sigma1 = lambda x: rotr(x,6) ^ rotr(x,11) ^ rotr(x,25)
ch = lambda e,f,g: (e&f)^(~e&g)&0xFFFFFFFF
maj = lambda a,b,c: (a&b)^(a&c)^(b&c)
Ho = [0x6a09e667, 0xbb67ae85, 0x3c6ef372, 0xa54ff53a, 0x510e527f,
0x9b05688c, 0x1f83d9ab, 0x5be0cd19]
T2_o_nop = (sigma0(Ho[0]) + maj(Ho[0], Ho[1], Ho[2])) % 2**32
assert hex(T2_o_nop) == '0x8909ae5' # 0x08909ae5
```

```
D_word = 4 D_bit = 6 T2^(o)_o = 0x08909ae5 residual ~ 16
```